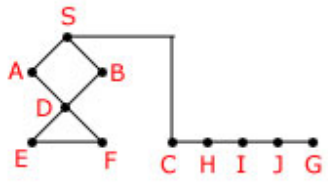
Tutorial #4: More Search (the Sequel)

We are trying to find a path from S to G in the following graph:




```
Connections (all undirected):
 S: A B C          F: D E
 A: S D            G: J
 B: S D            H: C I
 C: S H            I: H J
 D: A B E F        J: I G
 E: D F
```

The path length of all the links is 1, except for S-C which is of length 10.

The heuristic distance estimates to the goal are:

```
S = 10, A = 5, B = 11, C = 1, D = 7, E = 6,
F = 5, H = 3, I = 2, J = 1, G = 0
```

Assume that the implementation of Uniform Cost Search and A* places newly extended paths at the **front** of Q, with the new paths ordered by alphabetical order of the head state. The order on Q of paths will be used to resolve any ties in choosing the "best" path to pull off of Q, the path that appears **closer to the front** of Q is chosen.

Write the sequence of **states expanded** by the specified search methods in our implementation. In some searches, states may be repeated in the sequence. When using an expanded list, if the state pulled off Q is found in the expanded list, you should not include it in your answer (since its descendants are not generated). You will save yourself a LOT of confusion if you first convert the graph above into the appropriate search tree.

1. **Uniform Cost (NOT using Expanded List)**. This is tricky, please pay careful attention to the rule for settling ties given above. In each slot, enter a state name, including S and G:

S	A	B	D	D	B	E	F	A	E	F	E	F	E	F	C	H	I	J	G

2. **Uniform Cost (using strict Expanded List)**. In each slot, enter a state name, including S and G:

S	A	B	D	E	F	C	H	I	J	G

3. **A* (using strict Expanded List)**. For this answer, enter into each box both the state and the value of the f-value for the corresponding node. For example - S 10:

S10	A6	D9	F8	E9	C11	B12	H14	I14	J14	G14

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